Patent intelligence and patent landscapes

Transforming patent records into intelligence

Innovation Management · A.Y. 2026/27
Prof. Grazia Sveva Ascione · Tuesday 20 October 2026

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Today's lecture

01 **Patent intelligence: definitions**

02 **Patent landscapes: objectives, information, process**

03 **An example: the WIPO landscape of patents and the SDGs**

04 **Example: how to tag a SDG related patent**

01

Patent intelligence: definitions

From patent documents to decisions



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From patent data to patent intelligence

Four words that are often used as synonyms, but describe four different concepts.

Patent data

The **records** published by patent offices: bibliographic fields, text, classification codes, citations, legal events.

→

Patent information

Patent data organized **to answer a question**: who owns what, where it is protected, since when.

→

Patent analytics

Statistical and computational methods that transform patent data into actionable insights.

→

Patent intelligence

The use of patent analytics to support decisions on technologies, competitors, partners and markets.

more structure, more interpretation →

Patent intelligence is a part of **technology intelligence**: the systematic collection and analysis of information on technologies, here using patent data as the main source.

Why patents? A forward-looking source of data

What makes patent data interesting

- **Disclosure:** a patent is granted only if the invention is described in enough detail to be reproduced.
- **Timing:** applications are published (normally 18 months after the priority date), often years before the products reach the market.
- **Coverage:** almost 70% of technology data is captured in patent data, most of it publicly available.
- **Richness of information:** every document carries classification codes, dates, owners, inventors, citations and family links.

However..

- **Not every invention is patented:** secrecy and lead time are alternatives.
- **Propensity to patent differs across sectors**: electronics patents much more than, for instance, medical services.
- **Value is highly skewed:** a few patents hold most of the value.

For a data scientist: patent data are large, structured and free, but they measure inventive activity, not innovation itself. Every indicator is a modelling choice.

Key terms we will use today

Term	Definition
Patent family	All the applications filed in different countries to protect the same invention. Counting families avoids counting an invention twice.
DOCDB family	The EPO definition of family: documents that share exactly the same priorities.
Active patent family	A family with at least one pending published application or a granted patent not lapsed, withdrawn, invalidated or rejected.
International family	A family with protection sought beyond the home office.
IPC / CPC	International and Cooperative Patent Classification: hierarchical codes describing the technology of the invention.
NPL citation	A citation in a patent to non-patent literature, mostly scientific articles: a link between science and technology.
Freedom to operate (FTO)	A search to check that a product can be made and sold without infringing patents of others (blocking patents).

Patent intelligence guides decisions at every stage

An appropriate analysis of patent information gives guidance at every stage of IP and innovation management.

Analysis of technology trends	R&D prioritization and collaboration	IP protection strategies	Monitoring	Portfolio management and exploitation
Where is the technology going?	*What do we invest in, and with whom?*	*What, when and where do we file?*	*What are competitors filing? Are we free to operate?*	*What do we keep, license, sell or abandon?*

The cycle restarts: monitoring and portfolio decisions feed back into the next analysis of trends. Patent intelligence also supports its stakeholders by reducing the complexity of the IPR environment.

Who uses patent intelligence, and for what

Top management

- Overview of long-run technological trends
- Integrate market-based analyses of competitors' strategies

CTOs

- Relative strengths across the R&D portfolio
- Support to decisions on prioritization

Start-uppers

- Evaluation of potential collaborations
- Ex-ante freedom to operate

IPR managers

- IP filing and renewal approaches
- Licensing in and out opportunities
- Conveying information through the company

Researchers and field experts

- Awareness of specific innovation trends
- Identification of knowledge sources
- Limit the duplication of research effort

Same data, different questions: a landscape is designed around its user. Define the user before the search.

02

Patent landscapes: objectives, information, process

The main tool of patent intelligence

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What is a patent landscape?

Patent landscapes are **quantitative analyses**, combining both **technological** and **economic** perspectives, based on the processing of large-scale data on worldwide patent applications to derive information for **innovation management**.

Economic perspective

- Who invests in a technology, how much, and where.
- Competitors, partners, markets, value of portfolios.
- Questions of strategy and of policy.

Technological perspective

- What the technology is: classes, text, claims.
- How it evolves: trends, maturity, recombination.
- Where knowledge comes from: citations to patents and to science.

A landscape is a snapshot: it maps patent activity in a technology, for a set of actors or in a territory, at a given date.

Units of analysis and objectives

Units of analysis

- **Firm level:** a pool of competing firms.
- **Technology level:** a technological domain, a class of products or processes, a business solution or technical issue.
- **Geographical level:** a country, a region, a city.

Objectives of patent landscaping

- Understand on-going innovation trends.
- Benchmark the patenting strategies of competitors.
- Identify the key innovators in a technological domain.
- Discover blocking patents (a freedom-to-operate search).
- Find potential R&D partners, licensees or licensors.
- Other goals: evaluate R&D projects, organize a knowledge base, assess the impact of research proposals.

Source: Lecture 4, Technology intelligence through patent data.

Types of information: who, what, when, where, how

Question	Fields	Technique
Who	Applicants; Assignees (owners), inventors	The 'name game': the same firm or person is written in many ways. Entity disambiguation and consolidation of the ultimate owner.
What	IPC/CPC codes; title, abstract, claims, description	Codes are multi-label and hierarchical; text needs NLP. Both define the perimeter of a landscape.
When	Priority, filing and publication dates	The priority date is closest to the invention; publication comes about 18 months later (truncation at the end of series).
Where	Origin (priority office), extension markets (publication countries), inventor country of origin	Inventor address measures where R&D happens; offices measure where markets are.
How	Backward and forward citations, families, legal status events	Citation networks, family links, ownership changes.

You can add more data: firm financials, regional data, individual-level information, regulatory and historical events (e.g. COVID-19).

What a landscape measures

Innovation trends

- Applications or families by country
- Ranking of key innovators or sub-fields
- International extensions (family size)
- Patent density in the technology space

Competitors

- Competitor portfolios
- International extension (family size)
- Size and obsolescence (residual life)
- Collaborations (co-assignments)
- Specialization
- Transactions (reassignments, i.e. changes in patent ownership)

High value or high risk

- Citation profiles
- Prosecution history and litigation
- Science-based inventions (NPL references)
- Standard-essential patents

Data sources

Free online databases

- **TIP**: technology intelligence platform → PATSTAT: patents from all over the world for free
- **Espacenet** (EPO): worldwide.espacenet.com
- **PatentsView** (USPTO): accessible through USPTO website
- **UIBM** (Italian Patent and Trademark Office)
- **Google Patents:** patents.google.com
- **Lens.org:** lens.org (oriented towards analytics)

Commercial platforms

- **Orbit** (Questel): available at PoliTO
- **Derwent Innovation** (Clarivate)
- **PatentSight** (LexisNexis): the source of the WIPO report of today

Other databases (examples)

- **REGPAT** (OECD): patents linked to regions
- OECD patent quality dataset: set of patent quality metrics already calculated for EP and US patents
- Patent-to-paper citation datasey (Marx & Fuegi, 2020)

The process of a patent landscape

Complex information

- More than 140 million patent documents
- Additional sources: firm financials, regional data, individual-level information, regulatory and historical events

Elaboration

- Define the objective
- Define the perimeter through a search strategy

Output, easy to use

- Database
- Reporting

Strategies

- R&D
- Competitors
- Collaborations
- Markets

The search strategy is a classification problem. Every landscape starts by deciding which patents belong to the perimeter and which do not. When the perimeter is a technology, classification codes may help.

03

An example: the WIPO landscape of patents and the SDGs

WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals

Where is this report ? Ascione > PW > useful_material

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What are the Sustainable Development Goals (SDGs)?

The Sustainable Development Goals are 17 global objectives with 169 specific targets, adopted by the UN General Assembly in 2015 as Agenda 2030: a blueprint for peace and prosperity covering social, economic and environmental issues.

Social people

- 1 No poverty
- 2 Zero hunger
- 3 Good health and well-being
- 4 Quality education
- 5 Gender equality
- 7 Affordable and clean energy
- 11 Sustainable cities and communities
- 16 Peace, justice and strong institutions

Economic prosperity

- 8 Decent work and economic growth
- 9 Industry, innovation and infrastructure
- 10 Reduced inequalities
- 12 Responsible consumption and production

Environmental planet

- 6 Clean water and sanitation
- 13 Climate action
- 14 Life below water
- 15 Life on land

17 **Partnerships for the goals** the means of implementation, common to the three dimensions

Goals, targets and indicators are defined in UN documents: https://sdgs.un.org/goals

Why the SDGs matter for patent analysis

15%

of the SDG targets on track, halfway to 2030 (WIPO, 2024)

- **Technology is one of the levers:** energy, water, health, food, cities.
- **Defined by outcomes, not by technologies:** each goal has targets and indicators monitored by the UN.
- **Not every goal has a technology:** four goals (8, 10, 16, 17) concern socioeconomic issues, with no patentable technology area in the WIPO mapping.

8 **10** **16** **17** left out of the WIPO report by construction

Source: WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

Why map patents to the SDGs?

The policy question

- How much inventive effort goes to each goal, and is it growing?
- Which technologies, countries and organizations drive it?
- Where are the gaps, where more research, development and translation are needed?

The answer of the report

- **Patents are objective and forward-looking: they disclose R&D years before products.**
- **Active patents also show commitment: owners pay renewal fees to keep them alive.**
- The mapping gives a quantifiable measure of the intellectual capital invested in each goal.

15.2 M
active patent families worldwide

4.7 M
of them SDG-related (31.4%)

100
SDG-linked technology categories

13 of 17
SDGs mapped to patents

For strategy: decision-makers, policymakers and innovators can allocate resources and find partners where inventive contributions are most needed.

Source: WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

What could the problems of mapping patents to the SDGs be?

No explicit link

Legal and technical text, with no clear mention of the SDGs.

Hidden relevance

A patent can be useful for an SDG without mentioning anything related to it.

Unequal reach of technology

Not all SDGs can be improved by technical solutions in the same way. For instance, SDG 5 Gender equality?

Very broad goals

Some SDGs are very general, like SDG 9 Industry, innovation and infrastructure.

In which SDGs do we expect to see more patents?

Your turn: map patents to the SDGs

10 minutes ;)

Propose a solution to decide whether a patent is related to an SDG.

Keywords?

Embeddings?

LLMs?

Classification?
Clustering?

Preliminary question: how the report defines an SDG patent

UN documents: goals, targets, indicators, metadata, policy → **100 patentable technology categories** → **A search strategy for each technology** → **Patents linked to one or more SDGs**

Inside each search strategy

- **Classification codes:** IPC, CPC and Japanese F-terms.
- **Text:** English title, abstract, claims and description, with machine translation where no official translation exists.
- Strategies are tailored to each technology to cover the scope described by the SDGs.

What follows from this design

- Expert-built Boolean queries: transparent in principle, but not published in full, so not reproducible.
- A patent can belong to several technologies, and so to several SDGs (multi-label).
- Goals without a technology (8, 10, 16, 17) are left out by construction.

The strategy of the report: four design choices

1 · Data

- **LexisNexis PatentSight**: over 100 million patent documents
- DOCDB simple families
- Active families on 31 December 2023
- Ultimate **owner consolidated** (M&A, reassignments)

2 · Mapping

- 100 technology categories found in UN goals, targets, indicators, metadata and policy documents
- One search strategy per technology
- 13 of 17 SDGs mapped

3 · Breakdowns

- By **SDG**: counts, trends, shares
- By **technology**: WIPO concordance, 35 fields in 5 sectors (Schmoch, 2008)
- Maturity: the Innovation Maturity Matrix

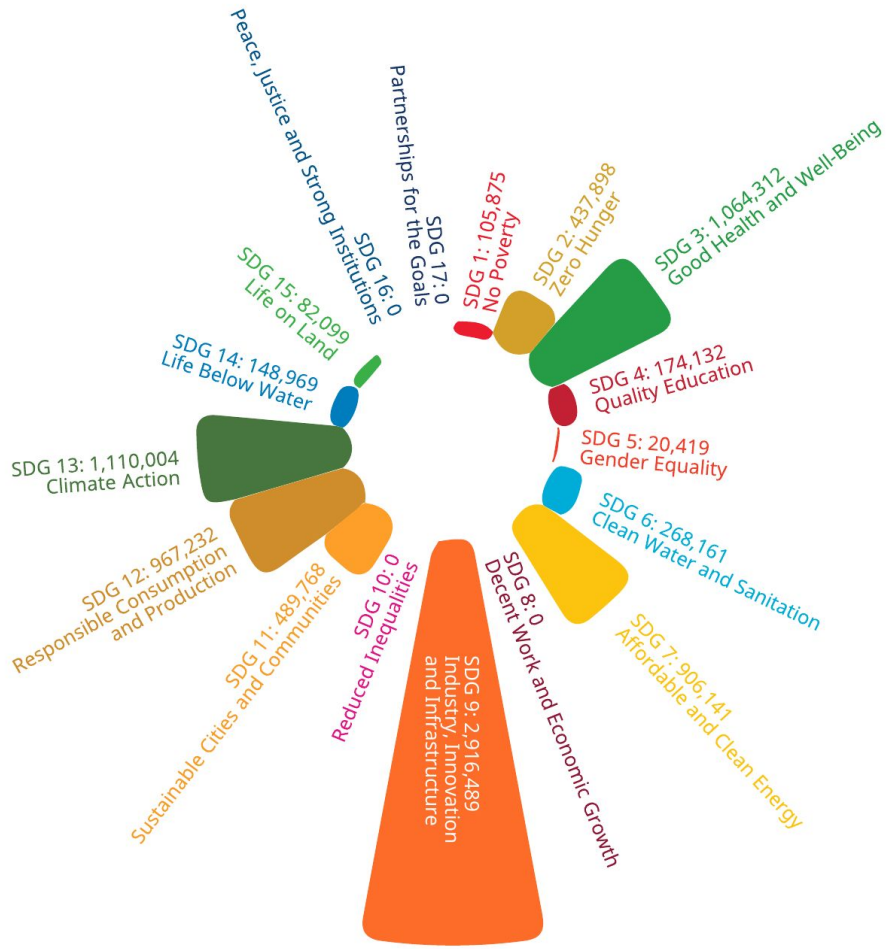
4 · Who and where

- **International families only**, to control for value and national biases
- Filing authorities and inventor location
- **Top 25 owners** per region and in academia

Read it as a template: data → perimeter → indicators → actors. It is the same sequence you will follow in the project work.

Source: WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0). Appendix A.1–A.4.

Almost one active patent family in three is SDG-related

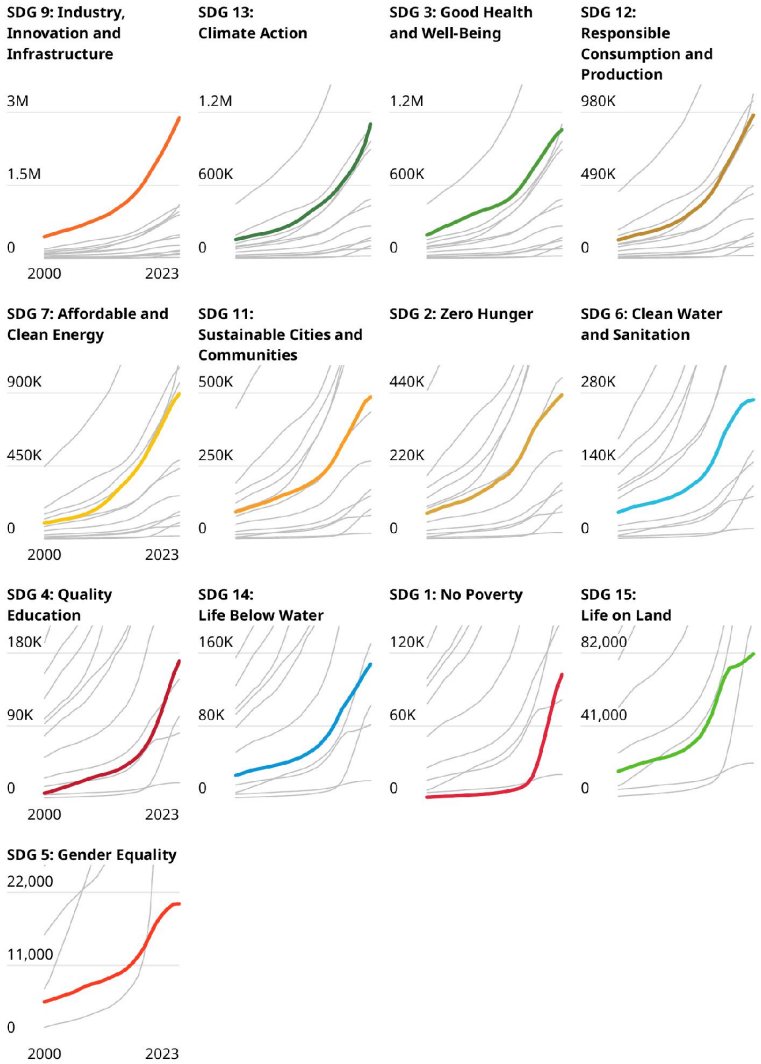


How to read it

- Each petal is an SDG; its length is the number of active patent families.
- 15.2 million active families worldwide; 4.7 million (31.4%) relate to the SDGs.
- **SDG 9 Industry, Innovation and Infrastructure** is by far the largest: 2,916,489 families.
- It covers electronics, manufacturing and materials, which are heavily patented.
- Then SDG 3 Health (1.06 M) and SDG 13 Climate Action (1.11 M).
- Families can be counted under more than one SDG.

Source: Figure 1 in WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

All SDGs grow, at different speeds

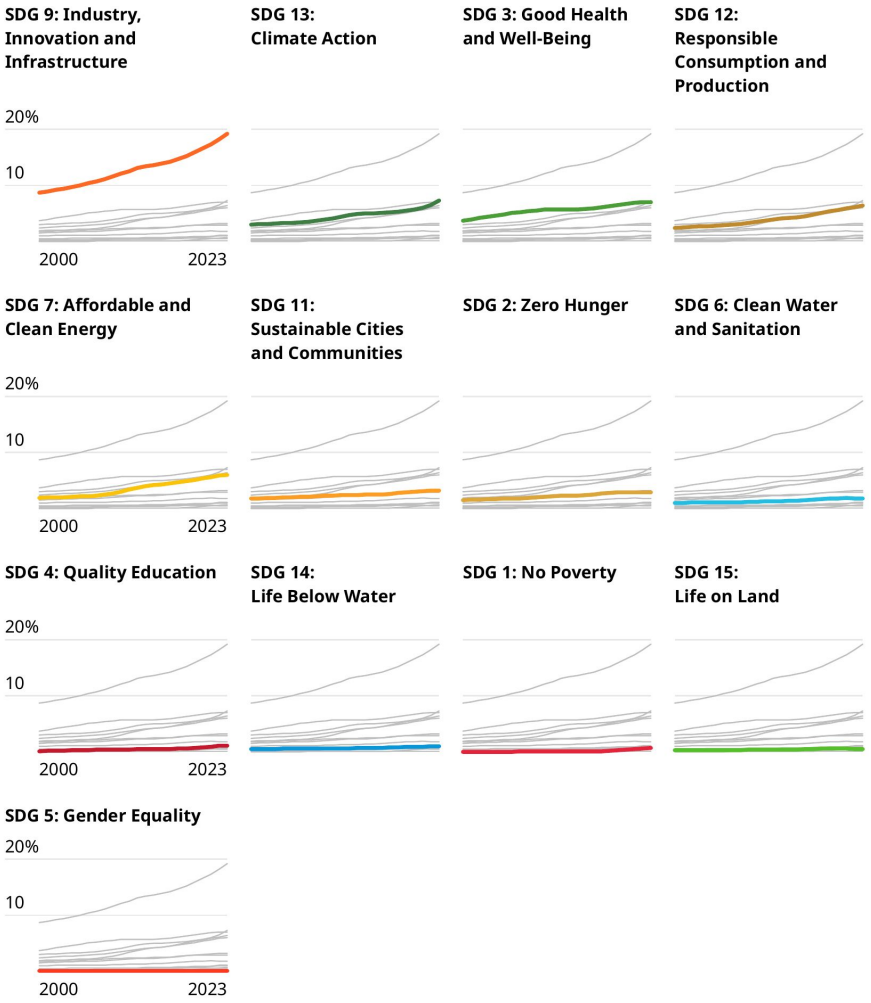


How to read it

- Small multiples, 2000–2023: the coloured line is the SDG, grey lines are the other SDGs.
- Each panel has its own scale: compare shapes, not levels.
- All mapped SDGs grow; SDG 13 and SDG 7 accelerate most.
- Small SDGs (5 Gender Equality, 1 No Poverty) grow fast from a low base.

Source: Figure 2 in WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

SDG patents grow faster than patents overall



How to read it

- Share of all active families attributed to each SDG.
- SDG 9 rises from under 10% to about 20% of all active patents.
- **SDG 13 Climate Action and SDG 7 Clean Energy** show stronger upward trends than most SDGs.
- A rising share means the SDG grows faster than total patenting, which is itself growing.
- **Why shares:** they control for the general growth of patent filings.

Source: Figure 3 in WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

The Innovation Maturity Matrix



Two dimensions

- **Innovation intensity (y, log scale):** number of published SDG-related applications.
- **Relative recency (x):** how recently they were filed, normalized so that the whole SDG dataset has recency 1.

$$R = \Sigma_i (w_i \cdot i) / (n \cdot \Sigma_i w_i)$$

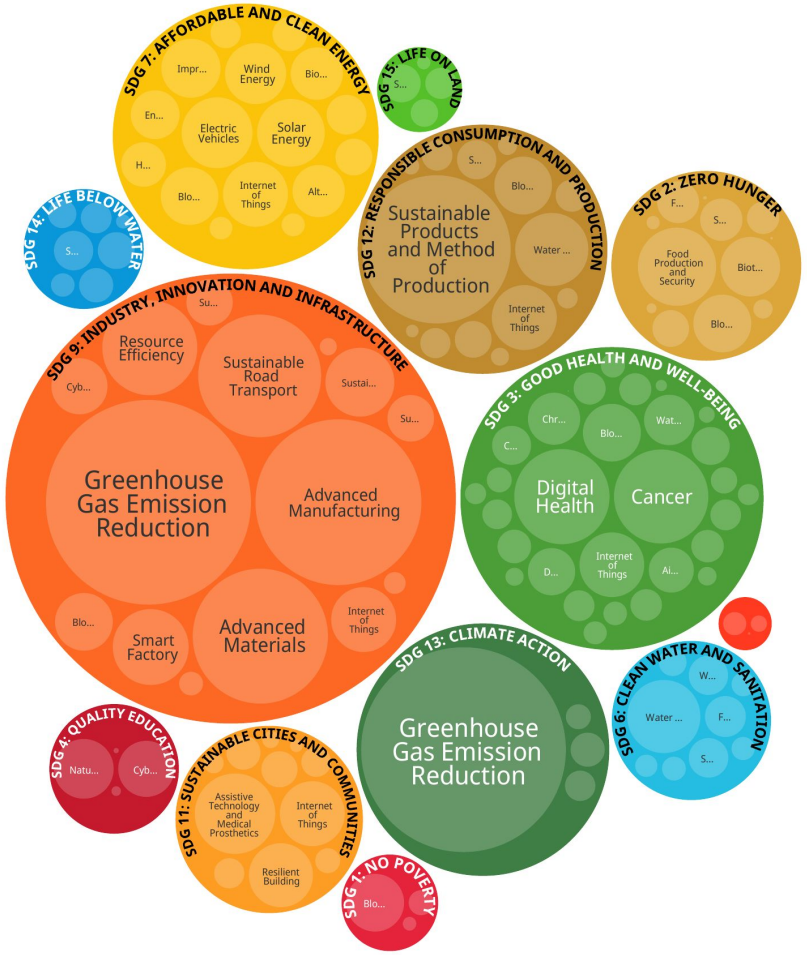
i = 1 for the first year, increasing by 1 each year; n = number of years; w_i = applications filed in year i. Relative recency = R of the SDG / R of all SDG patents.

Four quadrants

- **Current hot topics:** many and recent (SDG 9, 13, 7, 12).
- **Emerging interest:** few but recent (SDG 1, 4, 6, 14, 15).
- **Mature sectors** and modest development: older filings.

Source: Figure 4 and Appendix A.4 in WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0). Method by IPOS International.

One SDG, many technologies; one technology, many SDGs

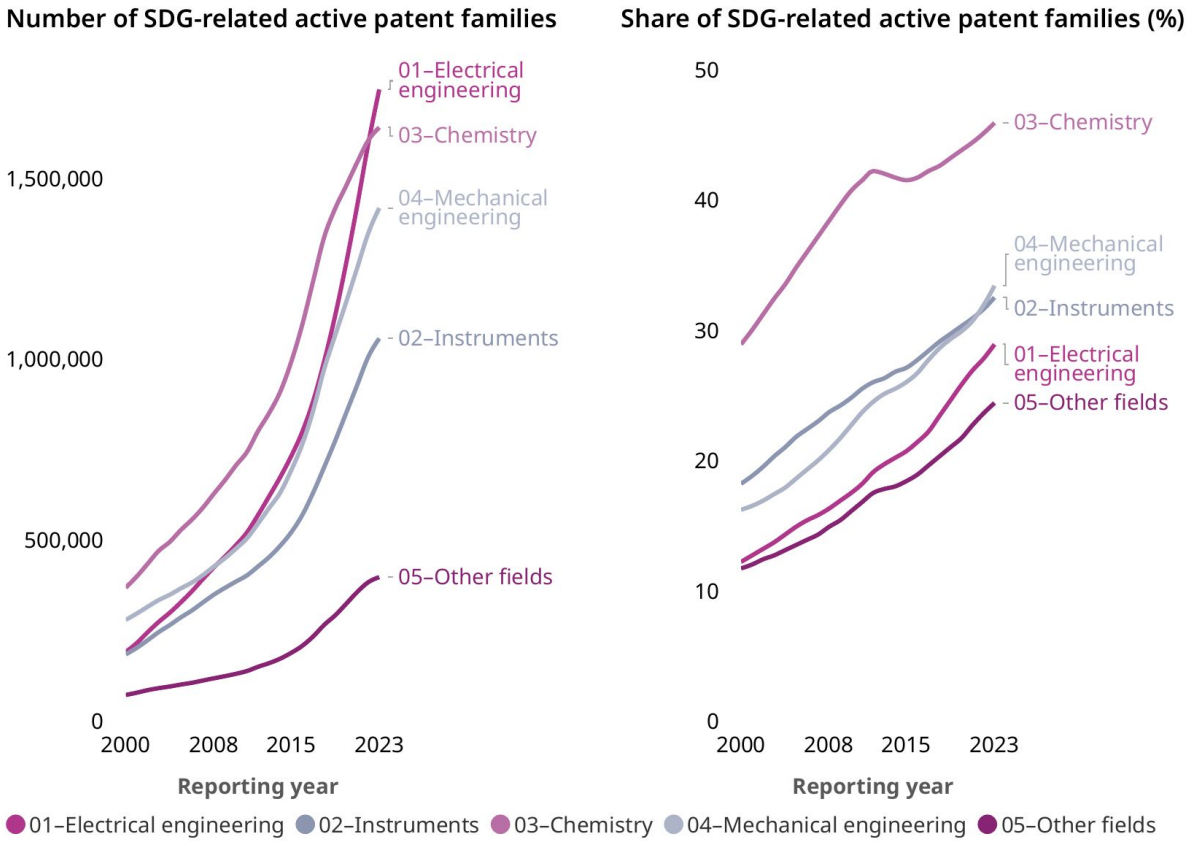


How to read it

- The 100 SDG-linked technologies, grouped by SDG; circle size is the number of active families.
- SDG 9 contains fewer but larger technology areas; SDG 3 many small, specific ones (e.g. cancer, hepatitis).
- **Overlaps:** Greenhouse Gas Emission Reduction appears in both SDG 9 and SDG 13.
- **For modelling:** SDG classification is multi-label by nature. Counts by SDG do not add up to the total.

Source: Figure 5 in WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

SDG patents across technology sectors

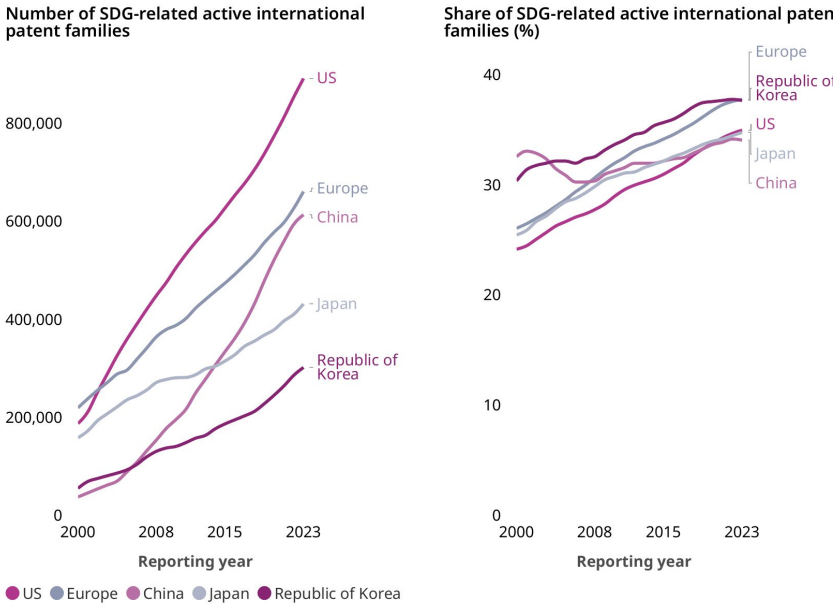


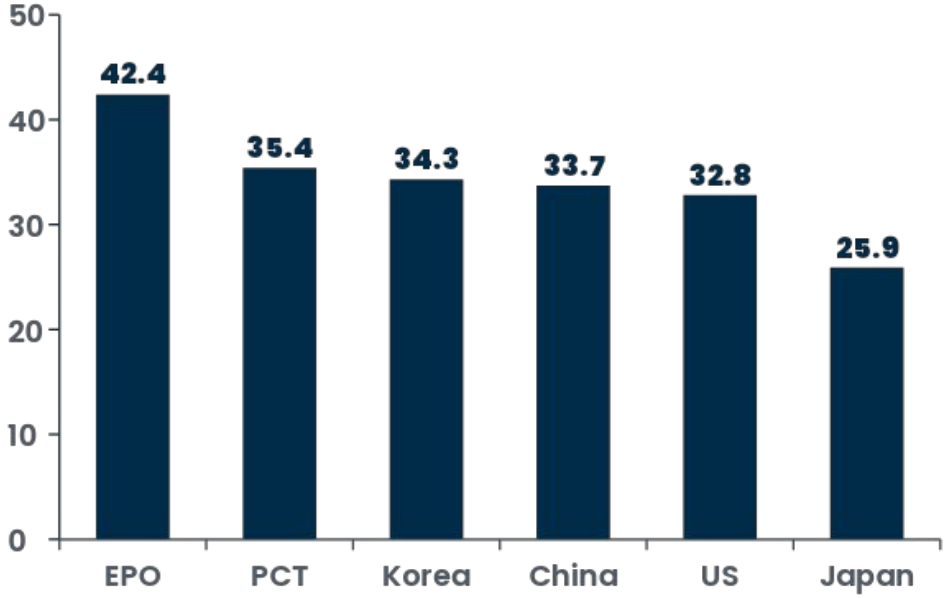
How to read it

- WIPO concordance: IPC codes are mapped to 35 fields grouped in 5 sectors.
- Left: number of SDG-related families; right: their share of each sector.
- **Chemistry dominates the share** (pharmaceuticals, processes for emission reduction).
- Electrical engineering surges at a faster pace in recent years.
- **Why this breakdown:** it connects the SDG view to the standard technology view of patent statistics.

Source: Figure 6 in WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

Where SDG inventions are protected





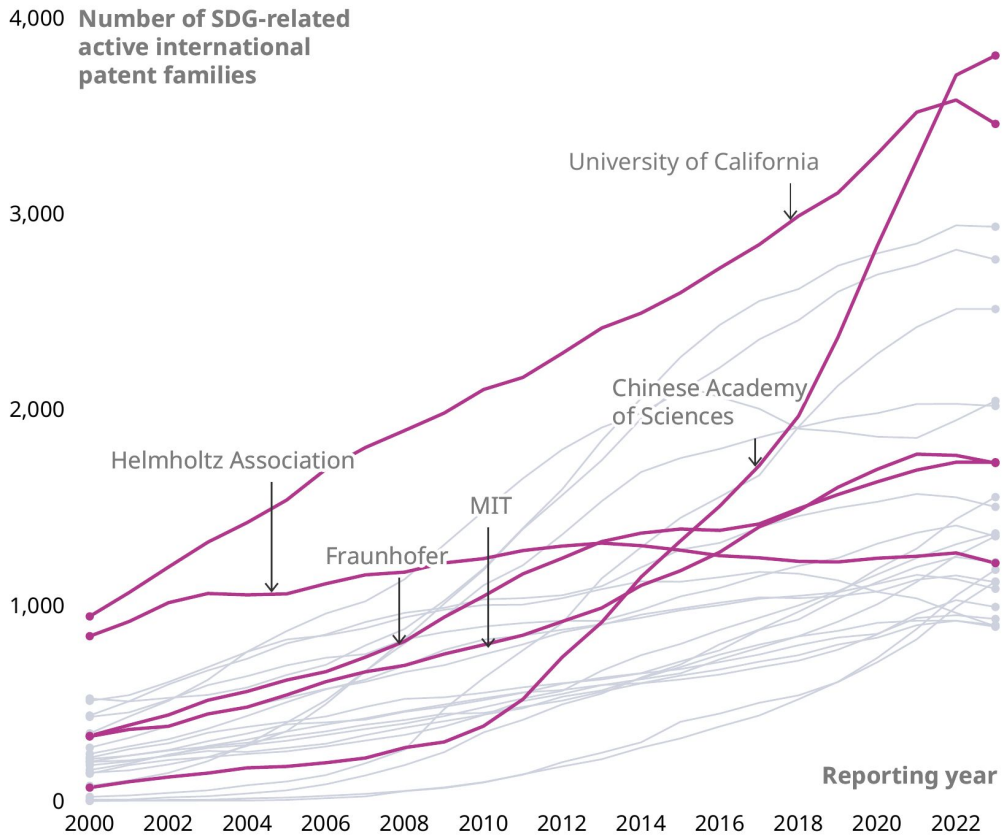
Share of active patents that are SDG-related, by office (%)

Why international families for this part of the analysis

- They keep only inventions their owners judged **worth protecting abroad: a proxy for quality and value**
- They reduce national biases: about 48% of all active families are filed in one authority only, roughly half of all families are China-only.

Source: Figure 11, Figure 26 and Appendix A.3 in WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

Who owns SDG patents: academia and research



Source: WIPO, based on patent data from PatentSight, January 2024.

How to read it

- International SDG families of the top 25 academic and research owners, 2000–2023.
- The University of California led for two decades; the Chinese Academy of Sciences now leads (3,805 families).
- Then UC (3,456), CEA (2,931), CNRS (2,764), Tsinghua (2,511), ETRI, ITRI, Fraunhofer, MIT (1,725).
- SDG patents are 68% of UC's portfolio and 64% of MIT's.
- Asian institutions spread their families more evenly across SDGs.

Source: Figure 23 and Table 6 in WIPO (2024), Mapping Innovations: Patents and the Sustainable Development Goals, based on LexisNexis PatentSight data, January 2024 (CC BY 4.0).

Reading the report as patent intelligence

Top management

Which SDGs are hot topics and which are emerging (maturity matrix).

CTOs

Which technology fields connect to which goals (concordance, Figure 10).

IPR managers

Where to file: SDG inventions favour the PCT and EPO routes.

Researchers and start-uppers

Who the key owners and potential partners are, by region and in academia.

What the report cannot tell us

- Whether an SDG-related patent has any real impact on the goal: it measures inventive effort, not outcomes.
- How each patent was assigned: the search strategies are proprietary, so the perimeter cannot be replicated or tested.
- Anything on SDGs 8, 10, 16 and 17.

The still open question: how do we decide, reproducibly and at scale, that a patent is related to an SDG?

04

How to decide that a patent is SDG-related

The techniques, and my solution. Based on Ascione & Tamagnone (2025), arxiv.org/abs/2509.09303



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EPO CodeFest Spring 2025: classifying patent data for the SDGs



The same problem, as a competition

- The EPO asked developers and data scientists to build an **automated system that classifies patent data by SDG**.
- **33 proposals** from teams and individuals in about 30 countries.

CodeFest 2025 page →

Source: EPO, CodeFest Spring 2025 on classifying patent data for sustainable development (epo.org).

The task: a non-technological classification

Patent-to-SDG classification is the task of learning a function $f(x) \rightarrow y \in [0, 1]^{17}$ that maps a patent x (its text and metadata) to its relevance for each of the 17 SDGs.

Why it is hard

- SDGs are defined by societal outcomes, not by technologies.
- IPC and CPC describe technologies: there is no SDG class.
- A patent can serve several goals: multi-label, soft.

The real bottleneck

- No gold standard: no large, public, labelled dataset of SDG patents.
- Without labels, supervised learning cannot be trained or even evaluated.
- Expert labelling is costly and needs judgement across disciplines.

Easier relatives

- **Technology classification:** IPC/CPC labels exist for every patent.
- **Critical raw materials, circular economy:** partly covered by CPC (e.g. Y02, Y10), combined with keywords.

The choice of technique is a choice about where the labels come from.

Source: Ascione & Tamagnone (2025), From scratch to silver, arXiv:2509.09303.

Four families of techniques in the literature

Approach	Studies	How it works	Main limitation
Keyword-based	Van der Waal et al. (2021); Ascione (2023)	SDG thesauri from Agenda 2030; expanded with TF–IDF or embeddings	Manual effort; no context; polysemy and synonyms
Transfer learning	Hajikhani & Suominen (2022)	Train on SDG-labelled scientific papers, apply to patents	Science and patent language differ; one SDG per patent
Supervised	Maehara et al. (2025)	Fine-tune a transformer on 8,000 Japanese patents labelled by the JPO	One jurisdiction; expensive labels; generalizability
Citation-based	Ciarli et al. (2022)	A patent citing an SDG-tagged article (NPL) gets its SDG	Citations are noisy or strategic; many patents cite no science

Plus the expert queries of commercial providers (the WIPO report): classification codes and keywords, built technology by technology.

Source: Ascione & Tamagnone (2025), From scratch to silver, arXiv:2509.09303. Table 1.

Technique 1: keyword and classification queries

How it works

- Build a dictionary of SDG terms from Agenda 2030 targets and indicators, often combined with IPC/CPC codes.
- Van der Waal et al. (2021): 'blue' patents for unmet social needs, 'green' patents for established sustainable technologies.
- Ascione (2023): TF–IDF and vector representations to expand the vocabulary.

An illustrative query

```
( CPC = Y02E10/5*  OR  CPC = H01L31/* )
OR  TI,AB,CL = ( "solar cell"  OR  photovoltaic* )
AND NOT  TI = ( "solar radiation sensor" )
```

→ **technology "solar energy"** → **SDG 7, SDG 13**

Strengths and weaknesses

- **Transparent and cheap to run:** every match can be explained.
- **But:** no context, synonyms are missed, words have several meanings. 'Rural' also retrieves CN109446992B, a remote-sensing AI system for military or real-estate uses. No accepted SDG patent dictionary exists.

Source: Ascione & Tamagnone (2025), From scratch to silver, arXiv:2509.09303. Section 2.1.

Techniques 2 and 3: learn from labeled examples

Transfer learning (Hajikhani & Suominen, 2022)

- Scientific publications already carry SDG labels (e.g. the Elsevier SDG queries).
- Train a classifier on papers, then apply it to European patents.
- **Domain shift:** patents are written to claim legal scope, papers to report findings. Vocabulary and structure differ.
- Assumes one SDG per patent, and is hard to validate on patents.

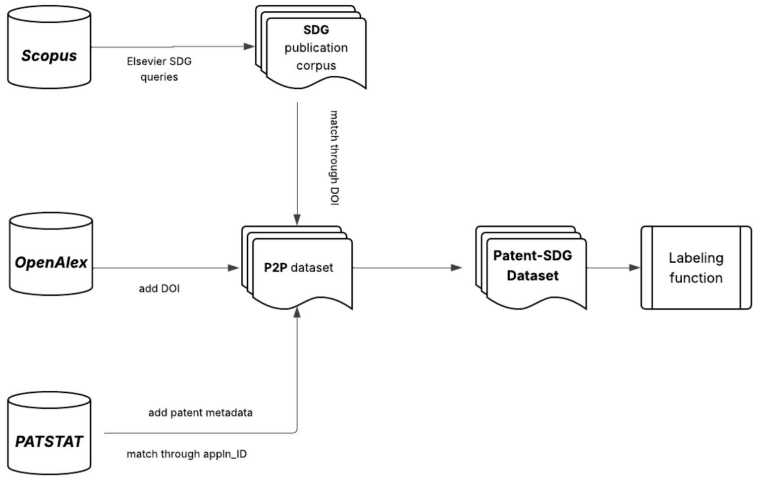
Supervised fine-tuning (Maehara et al., 2025)

- 8,000 Japanese patents labelled by the Japan Patent Office.
- Multi-stage fine-tuning of a patent-domain DeBERTa (a BERT-family transformer).
- Strong performance on its test set.
- **But:** labels from one office and one language; expensive expert annotation; unclear generalization.

The trade-off: transfer learning has many labels from the wrong domain; supervised learning has the right domain but few labels.

Source: Ascione & Tamagnone (2025), From scratch to silver, arXiv:2509.09303. Section 2.1.

Technique 4: let the citations label the patents



How it works (Ciarli et al., 2022)

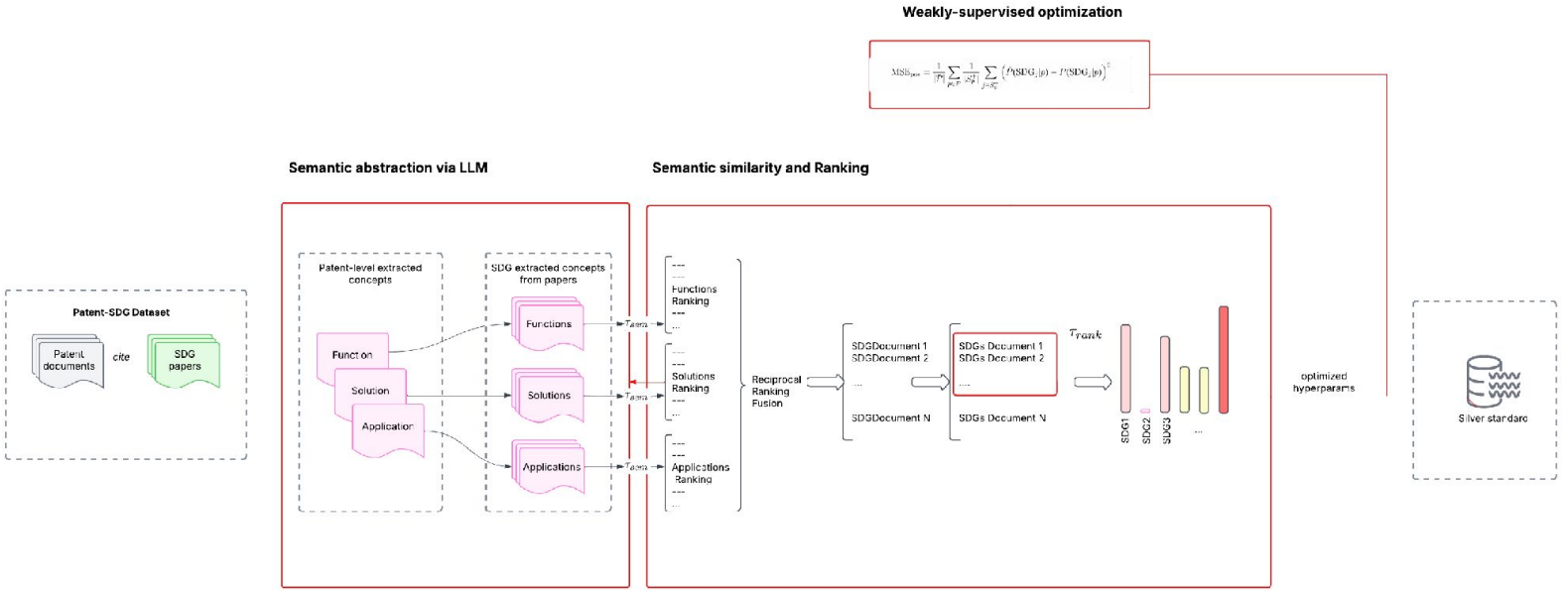
- Scientific articles are tagged with SDGs by Boolean queries (Elsevier SDG Research Mapping).
- Patents cite articles in their non-patent literature (NPL).
- A patent citing an SDG-tagged article inherits its SDG.

Why the signal is weak

- **Noisy:** front-page citations are often added by examiners, for legal reasons.
- **Incomplete:** coverage differs across fields; many SDG patents cite no science at all. NPL is 'not only noisy but highly incomplete' (Nagaoka & Yamauchi, 2015).

Source: Figure 1 (data pipeline) in Ascione & Tamagnone (2025), From scratch to silver, arXiv:2509.09303.

The labeling function: from noisy citations to silver labels



1 · Semantic abstraction

An LLM extracts functions, solutions and applications from patents and SDG papers.

2 · Similarity and ranking

Concepts are embedded and compared; rankings are fused (RRF) into a 17-dim SDG vector.

3 · Weakly supervised optimization

Thresholds and top-N are tuned to recover the NPL signal, without penalizing new labels.

Source: Figure 2 in Ascione & Tamagnone (2025), From scratch to silver, arXiv:2509.09303.

Questions?

Tomorrow
Landscaping tutorial 1: patent attributes · Wednesday 21 October, 13:00–14:30 · Room R3

Next lecture
Knowledge spillovers · Wednesday 28 October

Politecnico di Torino